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SHEAR THICKENING FLUID BASED ROTARY POWER SHUNT MECHANISM

Abstract

A power shunt for shunting rotary power from a load device includes a shear thickening fluid (STF) and a chamber containing the STF. The power shunt further includes a drive shaft housed radially within a drive side section of the chamber protruding outward from an end of the chamber for coupling to a lock preventing rotation of the drive shaft. The power shunt further includes a load shaft housed radially within a load side section of the chamber protruding outward through a shaft collar from another end of the chamber for coupling to the load device. The power shunt further includes a drive turbine housed radially within the drive side section coupled to the drive shaft. The power shunt further includes a load turbine housed radially within the load side section at an adjustable operational distance from the drive turbine and coupled to the load shaft.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present U.S. Utility patent application claims priority pursuant to 35 U.S.C. § 120 as a continuation of U.S. Utility application Ser. No. 17/703,253, entitled “SHEAR THICKENING FLUID BASED ROTARY POWER SHUNT MECHANISM” filed Mar. 24, 2022, issuing as U.S. Pat. No. 12,297,875 on May 13, 2025, which claims priority pursuant to 35 U.S.C. § 119 (e) to U.S. Provisional Application No. 63/321,871, entitled “SHEAR THICKENING FLUID BASED ROTARY POWER COUPLER AND SHUNT MECHANISM”, filed Mar. 21, 2022, expired, all of which are hereby incorporated herein by reference in their entirety and made part of the present U.S. Utility patent application for all purposes.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT [0002] Not Applicable.

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC [0003] Not Applicable.

BACKGROUND OF THE INVENTION

Technical Field of the Invention

[0004] This invention relates generally to systems that measure and control mechanical movement and more particularly to sensing and controlling of a linear and/or rotary movement mechanism that includes a chamber with dilatant fluid (e.g., a shear thickening fluid).

Description of Related Art

[0005] Many mechanical mechanisms are subject to undesired movement that can lead to annoying sounds, property damage and/or loss, and personal injury and even death. Desired and undesired movements of the mechanical mechanisms may involve a wide range of forces. A need exists to control the wide range of forces to solve these problems.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0006] FIG. 1A is a schematic block diagram of an embodiment of a mechanical and computing system in accordance with the present invention;

[0007] FIG. 1B is a graph of viscosity vs. shear rate for an aspect of an embodiment of a mechanical and computing system in accordance with the present invention;

[0008] FIG. 1C is a graph of plunger velocity vs. force applied to the plunger for an aspect of an embodiment of a mechanical and computing system in accordance with the present invention;

[0009] FIG. 2A is a schematic block diagram of an embodiment of a computing entity of a computing system in accordance with the present invention;

[0010] FIG. 2B is a schematic block diagram of an embodiment of a computing device of a computing system in accordance with the present invention;

[0011] FIG. 3 is a schematic block diagram of another embodiment of a computing device of a computing system in accordance with the present invention;

[0012] FIG. 4 is a schematic block diagram of an embodiment of an environment sensor module of a computing system in accordance with the present invention;

[0013] FIG. 5A is a cross-section diagram of an embodiment of a rotary power coupler in accordance with the present invention;

[0014] FIG. 5B is a diagram of an embodiment of a drive turbine in accordance with the present invention;

[0015] FIG. 5C is a diagram of an embodiment of a load turbine in accordance with the present invention;

[0016] FIGS. 5D-5E are graphs portraying embodiments of operation of a rotary power coupler in accordance with the present invention;

[0017] FIG. 6A is a cross-section diagram of an embodiment of another rotary power coupler in accordance with the present invention;

[0018] FIG. 6B is a graph portraying an embodiment of operation of another rotary power coupler in accordance with the present invention;

[0019] FIG. 6C is a cross-section diagram of another embodiment of a rotary power coupler in accordance with the present invention;

[0020] FIG. 6D is a graph portraying another embodiment of operation of a rotary power coupler in accordance with the present invention;

[0021] FIGS. 7A-7B are cross-section diagrams of an embodiment of another rotary power coupler illustrating an example of controlling coupling of rotary power in accordance with the present invention;

[0022] FIG. 7C is a timing diagram set portraying an embodiment of operation of another rotary power coupler in accordance with the present invention;

[0023] FIG. 8A is a cross-section diagram of an embodiment of a rotary power shunt in accordance with the present invention;

[0024] FIG. 8B is a graph portraying an embodiment of operation of a rotary power shunt in accordance with the present invention;

[0025] FIG. 9A is a cross-section diagram of an embodiment of another rotary power shunt in accordance with the present invention;

[0026] FIG. 9B is a graph portraying an embodiment of operation of another rotary power shunt in accordance with the present invention;

[0027] FIGS. 10A-10B are cross-section diagrams of an embodiment of another rotary power shunt illustrating an example of controlling shunting of rotary power in accordance with the present invention; and

[0028] FIG. 10C is a timing diagram set portraying an embodiment of operation of another rotary power shunt in accordance with the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0029] FIG. 1A is a schematic block diagram of an embodiment of a mechanical and computing system that includes a set of head units **10-1** through **10-N**, objects **12-1** through **12-3**, computing entities **20-1** through **20-N** associated with the head units **10-1** through **10-N**, and a computing entity **22**. The objects include any object that has mass and moves. Examples of an object include a door, an aircraft wing, a portion of a building support mechanism, and a particular drivetrain, etc.

[0030] The cross-sectional view of FIG. 1A illustrates a head unit that includes a chamber **16**, a piston **36**, a plunger **28**, a plunger bushing **32**, and a chamber bypass **40**. The chamber **16** contains a shear thickening fluid (STF) **42**. The chamber **16** includes a back channel **24** and a front channel **26**, where the piston partitions the back channel **24** and the front channel **26**. The piston **36** travels axially within the chamber **16**. The chamber **16** may be a cylinder or any other shape that enables movement of the piston **36** and compression of the STF **42**. The STF **42** is discussed in greater detail with reference to FIGS. 1B and 1C.

[0031] The plunger bushing **32** guides the plunger **28** into the chamber **16** in response to force from the object **12-1**. The plunger bushing **32** facilitates containment of the STF within the chamber **16**. The plunger bushing **32** remains in a fixed position relative to the chamber **16** when the force from the object moves the piston **36** within the chamber **16**. In an embodiment the plunger bushing **32** includes an O-ring between the plunger bushing **32** and the chamber **16**. In another embodiment the plunger bushing **32** includes an O-ring between the plunger bushing **32** and the plunger **28**.

[0032] The piston **36** includes a piston bypass **38** between opposite sides of the piston to facilitate flow of a portion of the STF between the opposite sides of the piston (e.g., between the back channel **24** and the front channel **26**) when the piston travels through the chamber in an inward or an outward direction.

[0033] Alternatively, or in addition to, the chamber bypass **40** is configured between opposite ends of the chamber **16**, wherein the chamber bypass **40** facilitates flow of a portion of the STF between the opposite ends of the chamber when the piston travels through the chamber in the inward or outward direction (e.g., between the back channel **24** and the front channel **26**).

[0034] In alternative embodiments, the piston bypass **38** and the chamber bypass **40** includes mechanisms to enable STF flow in one direction and not an opposite direction. In further alternative embodiments, a control valve within the piston bypass **38** and/or the chamber bypass controls the STF flow between the back channel **24** and the front channel **26**. Each bypass includes one or more of a one-way check valve and a variable flow valve.

[0035] The plunger **28** is operably coupled to a corresponding object by one of a variety of approaches. A first approach includes a direct connection of the plunger **28** to the object **12-1** such that linear motion in any direction couples from the object **12-1** to the plunger **28**. A second approach includes the plunger **28** coupled to a cap **44** which receives a one way force from a strike **48** attached to the object **12-2**. A third approach includes a pushcap **46** that receives a force from a rotary-to-linear motion conversion component that is attached to the object **12-3**. In an example, the object **12-3** is connected to a camshaft **110** which turns a cam **109** to strike the pushcap **46**.

[0036] In an embodiment, two or more of the head units are coupled by a head unit connector **112**. When so connected, actuation of a piston in a first head unit is essentially replicated in a piston of a second head unit. The head unit connector **112** includes a mechanical element between plungers of the two or more head units and/or direct connection of two or more plungers to a common object. For example, plunger **28** of head unit **10-1** and plunger **28** of head unit **10-2** are directly connected to object **12-1** when utilizing a direct connection.

[0037] Further associated with each head unit is a set of emitters and a set of sensors. For example, head unit **10-N** includes a set of emitters **114-N-1** through **114-N-M** and a set of sensors **116-N-1** through **116-N-M**. Emitters includes any type of energy and or field emitting device to affect the STF, either directly or indirectly via other nanoparticles suspended in the STF. Examples of emitter categories include light, audio, electric field, magnetic field, wireless field, etc. Specific examples of fluid manipulation emitters include a variable flow valve associated with a bypass or injector or similar, a mechanical vibration generator, an image generator, a light emitter, an audio transducer, a speaker, an ultrasonic sound transducer, an electric field generator, a magnetic field generator, and a radio frequency wireless field transmitter. Specific examples of magnetic field emitters include a Helmholtz coil, a Maxwell coil, a permanent magnet, a solenoid, a superconducting electromagnet, and a radio frequency transmitting coil.

[0038] Sensors include any type of energy and/or field sensing device to output a signal that represents a reaction, motion or position of the STF. Examples of sensor categories include bypass valve position, mechanical position, image, light, audio, electric field, magnetic field, wireless field, etc. Specific examples of fluid flow sensors include a valve opening detector associated with the chamber **16** or any type of bypass (e.g., piston bypass **38**, chamber bypass **40**, a reservoir injector, or similar), a mechanical position sensor, an image sensor, a light sensor, an audio sensor, a microphone, an ultrasonic sound sensor, an electric field sensor, a magnetic field sensor, and a radio frequency wireless field sensor. Specific examples of magnetic field sensors include a Hall effect sensor, a magnetic coil, a rotating coil magnetometer, an inductive pickup coil, an optical magnetometry sensor, a nuclear magnetic resonance sensor, and a caesium vapor magnetometer.

[0039] The computing entities **20-1** through **20-N** are discussed in detail with reference to FIG. **2A**. The computing entity **22** includes a control module **30** and a chamber database **34** to facilitate storage of history of operation, desired operations, and other aspects of the system.

[0040] In an example of operation, the head unit **10-1** controls motion of the object **12-1** and includes the chamber **16** filled at least in part with the shear thickening fluid **42**, the piston **36** housed at least partially radially within the chamber **16**, and the piston **36** is configured to exert pressure against the shear thickening fluid **42** in response to movement of the piston **36** from a force applied to the piston from the object **12-1**. The movement of the piston **36** includes one of traveling through the chamber **16** in an inward direction or traveling through the chamber **16** in an outward direction. The STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates.

[0041] The shear thickening fluid **42** (e.g., dilatant non-Newtonian fluid) has nanoparticles of a specific dimension that are mixed in a carrier fluid or solvent. Force applied to the shear thickening fluid **42** results in these nanoparticles stacking up, thus stiffening and acting more like a solid than a flowable liquid when a shear threshold is reached. In particular, viscosity of the shear thickening fluid **42** rises significantly when shear rate is increased to a point of the shear threshold. The relationship between viscosity and shear rates is discussed in greater detail with reference to FIGS. **1A** and **1B**.

[0042] In another example of operation, the object **12-1** applies an inward motion force on the plunger **28** which moves the piston **36** in words within the chamber **16**. As the piston moves inward, shear rate of the shear thickening fluid **42** changes. A sensor **116-1-1** associated with the chamber **16** of the head unit **10-1** outputs chamber I/O **160** to the computing entity **20-1**, where the chamber I/O **160** includes a movement data associated with the STF **42** as a result of the piston **36** moving inwards. Having received the chamber I/O **160**, the computing entity **20-1** interprets the chamber I/O **160** to reproduce the movement data.

[0043] The computing entity **20-1** outputs the movement data as a system message **162** to the computing entity **22**. The control module **30** stores the movement data in the chamber database **34** and interprets the movement data to determine whether to dynamically adjust the viscosity of the shear thickening fluid. Dynamic adjustment of the viscosity results in dynamic control of the movement of the piston **36**, the plunger **28**, and ultimately the object **12-1**. Adjustment of the viscosity affects velocity, acceleration, and position of the piston **36**.

[0044] The control module **30** determines whether to adjust the viscosity based on one or more desired controls of the object **12-1**. The desired controls include accelerating, deaccelerating, abruptly stopping, continuing on a current trajectory, continuing at a constant velocity, or any other movement control. For example, the control module **30** determines to abruptly stop the movement of the object **12-1** when the object **12-1** is a door and the door is detected to be closing at a rate above a maximum closing rate threshold level and when the expected shear rate versus viscosity of the shear thickening fluid **42** requires modification (e.g., boost the viscosity now to slow the door from closing too quickly).

[0045] When determining to modify the viscosity, the control module **30** outputs a system message

162 to the computing entity **20-1**, where the system message **162** includes instructions to immediately boost the viscosity beyond the expected shear rate versus viscosity of the shear thickening fluid **42**. Alternatively, the system message **162** includes specific information on the relationship of viscosity versus shear rate.

[0046] Having received the system message **162**, the computing entity **20-1** determines a set of adjustments to make with regards to the shear thickening fluid **42** within the chamber **16**. The set of adjustments includes one or more of adjusting STF **42** flow through the chamber bypass **40**, adjusting STF **42** flow through the piston bypass **38**, and activating an emitter of a set of emitters **114-1-1** through **114-N-1**. The flow adjustments include regulating within a flow range, stopping, starting, and allowing in one particular direction. For example, the computing entity **20-1** determines to activate emitter **114-1-1** to produce a magnetic field such as to interact with magnetic nanoparticles within the STF **42** to raise the viscosity. The computing entity **20-1** issues another chamber I/O **160** to the emitter **114-1-1** to initiate a magnetic influence process to boost the viscosity of the STF **42**.

[0047] In an alternative embodiment, the computing entity **22** issues another system message **162** to two or more computing entities (e.g., **20-1** and **20-2**) to boost the viscosity for corresponding head units **10-1** and **10-2** when the head unit connector **112** connects head units **10-1** and **10-2** and both head units are controlling the motion of the object **12-1**. For instance, one of the head units informs the computing entity **22** that the object **12-1** is moving too quickly inward and the predicted stopping power of the expected viscosity versus shear rate of the STF **42** of the head unit, even when boosted, will not be enough to slow the object **12-1** to a desired velocity or position. When informed that one head unit, even with a modified viscosity, is not enough to control the object **12-1**, the control module **30** determines how many other head units (e.g., connected via the head unit connector **112**) to apply and to dynamically modify the viscosity.

[0048] In yet another alternative embodiment, the computing entity **22** issues a series of system messages **162** to a set of computing entities associated with a corresponding set of head units to produce a cascading effect of altering of the viscosity of the STF **42** of each of the chambers **16** associated with the set of head units. For example, 3 head units are controlled by 3 corresponding computing entities to adjust viscosity in a time cascaded manner. For instance, head unit **10-1** abruptly changes the viscosity to attempt to slow the object **12-1** followed seconds later by head unit **10-2** abruptly changing the viscosity to attempt to further slow the object **12-1**, followed seconds later by head unit **12-3** abruptly changing the viscosity to attempt to further slow the object **12-1**.

[0049] In a still further alternative embodiment, the computing entity **22** conditionally issues each message of the series of system messages **162** to the set of computing entities associated with the corresponding set of head units to produce the cascading effect of altering of the viscosity of the STF **42** of each of the chambers **16** associated with the set of head units only when a most recent adaptation of viscosity is not enough to slow the object **12-1** with desired results. For example, the 3 head units are controlled by the 3 corresponding computing entities to adjust viscosity in a conditional time cascaded manner. For instance, head unit **10-1** abruptly changes the viscosity to attempt to slow the object **12-1** followed seconds later by head unit **10-2** abruptly changing the viscosity if head unit **10-1** was unsuccessful to attempt to further slow the object **12-1**, followed seconds later by head unit **12-3** abruptly changing the viscosity if head unit **10-2** was unsuccessful to attempt to further slow the object **12-1**.

[0050] FIG. **1B** is a graph of viscosity vs. shear rate for an aspect of an embodiment of a mechanical and computing system that includes a chamber, a shear thickening fluid, and a piston that moves through the chamber applying forces on the shear thickening fluid. The shear thickening fluid includes a non-Newtonian fluid since the relationship between shear rate and viscosity is nonlinear.

[0051] A relationship between compressive impulse (e.g., shear rate) and the viscosity of the shear

thickening fluid is nonlinear and may comprise one or more inflection points as the piston travels within the chamber in response to different magnitudes of forces and different accelerations. The viscosity of the STF may also be a function of other influences, such as electric fields, acoustical waves, magnetic fields, and other similar influences. As a first example of a response of a shear thickening fluid, a first range of shear rates in zone A has a decreasing viscosity as the shear rate increases and then in a second range of shear rates in zone B the viscosity increases abruptly. As a second example of a response of a diluted shear thickening fluid, the first range of shear rates in zone A extends to a higher level of shear rates with the decreasing viscosity and then in the still higher second range of shear rates in zone B the viscosity increases abruptly similar to that of the shear thickening include.

[0052] The shear thickening fluid includes particles within a solvent. Examples of particles of the shear thickening fluid include oxides, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, or a mixture thereof. Further examples of the particles of the shear thickening fluid include SiO₂, polystyrene, or polymethylmethacrylate.

[0053] The particles are suspended in a solvent. Example components of the solvent include water, a salt, a surfactant, and a polymer. Further example components of the solvent include ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone or a mixture thereof. Example particle diameters range from less than 100 μm to less than 1 millimeter. In an instance, the shear thickening fluid is made of silica particles suspended in polyethylene glycol at a volume fraction of approximately 0.57 with the silica particles having an average particle diameter of approximately 446 nm. As a result, the shear thickening fluid exhibits a shear thickening transition at a shear rate of approximately 102-103 s⁻¹.

[0054] A volume fraction of particles dispersed within the solvent distinguishes the viscosity versus shear rate of different shear thickening fluids. The viscosity of the STF changes in response to the applied shear stress. At rest and under weak applied shear stress, a STF may have a fairly constant or even slightly decreasing viscosity because the random distribution of particles causes the particles to frequently collide. However, as a greater shear stress is applied so that the shear rate increases, the particles flow in a more streamlined manner. However, as an even greater shear stress is applied so that the shear rate increases further, a hydrodynamic coupling between the particles may overcome the interparticle forces responsible for Brownian motion. The particles may be driven closer together, and the microstructure of the colloidal dispersion may change, so that particles cluster together in hydroclusters.

[0055] The viscosity curve of the STF can be fine-tuned through changes in the characteristics of the particles suspended in the solvent. For example, the particles shape, surface chemistry, ionic strength, and size affect the various interparticle forces involved, as does the properties of the solvent. However, in general, hydrodynamic forces dominate at a high shear stress, which also makes the addition of a polymer attached to the particle surface effective in limiting clumping in hydroclusters. Various factors influence this clumping behavior, including, fluid slip, adsorbed ions, surfactants, polymers, surface roughness, graft density (e.g., of a grafted polymer), molecular weight, and solvent, so that the onset of shear thickening can be modified. In general, the onset of shear thickening can be slowed by the introduction of techniques to prevent the clumping of particles. For example, influencing the STF with emissions from an emitter in proximal location to the chamber.

[0056] FIG. 1C is a graph of piston velocity vs. force applied to the piston for an aspect of an embodiment of a mechanical and computing system that includes a chamber, a shear thickening fluid, and a piston that moves through the chamber applying forces on the shear thickening fluid. The shear thickening fluid includes a non-Newtonian fluid since the relationship between shear rate and viscosity is nonlinear.

[0057] An example curve for a shear thickening fluid indicates that as more force is applied to the piston in zone A, a higher piston velocity is realized until the corresponding transition to zone B

occurs where the shear threshold affect takes hold and the viscosity abruptly increases significantly. When the viscosity increases abruptly, the piston velocity slows back down and may even stop. [0058] Another example curve for a diluted shear thickening fluid indicates that as more force is applied to the piston in zone A, an even higher piston velocity is realized until the corresponding transition to zone B occurs where the shear threshold affect takes hold and the viscosity abruptly increases significantly. When the viscosity increases abruptly, the piston velocity slows back down and may even stop.

[0059] FIG. 2A is a schematic block diagram of an embodiment of the computing entity (e.g., **20-1** through **20-N**; and **22**) of the mechanical and computing system of FIG. 1. The computing entity includes one or more computing devices **100-1** through **100-N**. A computing device is any electronic device that communicates data, processes data, represents data (e.g., user interface) and/or stores data.

[0060] Computing devices include portable computing devices and fixed computing devices. Examples of portable computing devices include an embedded controller, a smart sensor, a social networking device, a gaming device, a smart phone, a laptop computer, a tablet computer, a video game controller, and/or any other portable device that includes a computing core. Examples of fixed computing devices includes a personal computer, a computer server, a cable set-top box, a fixed display device, an appliance, and industrial controller, a video game counsel, a home entertainment controller, a critical infrastructure controller, and/or any type of home, office or cloud computing equipment that includes a computing core.

[0061] FIG. 2B is a schematic block diagram of an embodiment of a computing device (e.g., **100-1** through **100-N**) of the computing entity of FIG. 2A that includes one or more computing cores **52-1** through **52-N**, a memory module **102**, a human interface module **18**, an environment sensor module **14**, and an input/output (I/O) module **104**. In alternative embodiments, the human interface module **18**, the environment sensor module **14**, the I/O module **104**, and the memory module **102** may be standalone (e.g., external to the computing device). An embodiment of the computing device is discussed in greater detail with reference to FIG. 3.

[0062] FIG. 3 is a schematic block diagram of another embodiment of the computing device **100-1** of the mechanical and computing system of FIG. 1 that includes the human interface module **18**, the environment sensor module **14**, the computing core **52-1**, the memory module **102**, and the I/O module **104**. The human interface module **18** includes one or more visual output devices **74** (e.g., video graphics display, 3-D viewer, touchscreen, LED, etc.), one or more visual input devices **80** (e.g., a still image camera, a video camera, a 3-D video camera, photocell, etc.), and one or more audio output devices **78** (e.g., speaker(s), headphone jack, a motor, etc.). The human interface module **18** further includes one or more user input devices **76** (e.g., keypad, keyboard, touchscreen, voice to text, a push button, a microphone, a card reader, a door position switch, a biometric input device, etc.) and one or more motion output devices **106** (e.g., servos, motors, lifts, pumps, actuators, anything to get real-world objects to move).

[0063] The computing core **52-1** includes a video graphics module **54**, one or more processing modules **50-1** through **50-N**, a memory controller **56**, one or more main memories **58-1** through **58-N** (e.g., RAM), one or more input/output (I/O) device interface modules **62**, an input/output (I/O) controller **60**, and a peripheral interface **64**. A processing module is as defined at the end of the detailed description.

[0064] The memory module **102** includes a memory interface module **70** and one or more memory devices, including flash memory devices **92**, hard drive (HD) memory **94**, solid state (SS) memory **96**, and cloud memory **98**. The cloud memory **98** includes an on-line storage system and an on-line backup system.

[0065] The I/O module **104** includes a network interface module **72**, a peripheral device interface module **68**, and a universal serial bus (USB) interface module **66**. Each of the I/O device interface module **62**, the peripheral interface **64**, the memory interface module **70**, the network interface

module **72**, the peripheral device interface module **68**, and the USB interface modules **66** includes a combination of hardware (e.g., connectors, wiring, etc.) and operational instructions stored on memory (e.g., driver software) that are executed by one or more of the processing modules **50-1** through **50-N** and/or a processing circuit within the particular module.

[0066] The I/O module **104** further includes one or more wireless location modems **84** (e.g., global positioning satellite (GPS), Wi-Fi, angle of arrival, time difference of arrival, signal strength, dedicated wireless location, etc.) and one or more wireless communication modems **86** (e.g., a cellular network transceiver, a wireless data network transceiver, a Wi-Fi transceiver, a Bluetooth transceiver, a 315 MHz transceiver, a zig bee transceiver, a 60 GHz transceiver, etc.). The I/O module **104** further includes a telco interface **108** (e.g., to interface to a public switched telephone network), a wired local area network (LAN) **88** (e.g., optical, electrical), and a wired wide area network (WAN) **90** (e.g., optical, electrical). The I/O module **104** further includes one or more peripheral devices (e.g., peripheral devices **1-P**) and one or more universal serial bus (USB) devices (USB devices **1-U**). In other embodiments, the computing device **100-1** may include more or less devices and modules than shown in this example embodiment.

[0067] FIG. **4** is a schematic block diagram of an embodiment of the environment sensor module **14** of the computing device of FIG. **2B** that includes a sensor interface module **120** to output environment sensor information **150** based on information communicated with a set of sensors. The set of sensors includes a visual sensor **122** (e.g., to the camera, 3-D camera, 360° view camera, a camera array, an optical spectrometer, etc.) and an audio sensor **124** (e.g., a microphone, a microphone array). The set of sensors further includes a motion sensor **126** (e.g., a solid-state Gyro, a vibration detector, a laser motion detector) and a position sensor **128** (e.g., a Hall effect sensor, an image detector, a GPS receiver, a radar system).

[0068] The set of sensors further includes a scanning sensor **130** (e.g., CAT scan, MRI, x-ray, ultrasound, radio scatter, particle detector, laser measure, further radar) and a temperature sensor **132** (e.g., thermometer, thermal coupler). The set of sensors further includes a humidity sensor **134** (resistance based, capacitance based) and an altitude sensor **136** (e.g., pressure based, GPS-based, laser-based).

[0069] The set of sensors further includes a biosensor **138** (e.g., enzyme, microbial) and a chemical sensor **140** (e.g., mass spectrometer, gas, polymer). The set of sensors further includes a magnetic sensor **142** (e.g., Hall effect, piezo electric, coil, magnetic tunnel junction) and any generic sensor **144** (e.g., including a hybrid combination of two or more of the other sensors).

[0070] FIG. **5A** is a cross-section diagram of an embodiment of a rotary power coupler **200** that couples rotary power from a rotary power device **202** to a load device **204**. The rotary power device **202** includes one or more of a variety of power sources such as a fossil fuel engine, a hydro-turbine, a steam turbine, an electric motor, a servo motor, and a combination of any of the above. The load device **204** includes any type of device powered by and/or using rotary motion such as a vehicle, a wheel, an elevator, a conveyor system, an automatic door, etc.

[0071] The power coupler **200** includes a chamber **206**, a driveshaft **208**, a drive turbine **210**, a load turbine **212**, a load shaft **214**, and a cartridge seal **216**. The chamber **206** has an interior cylindrical shape and either an external cylindrical shape or a hexagonal external shape. In a primary embodiment the chamber **206** is affixed to a nonmoving structure associated with at least one of the rotary power device **202**, the load device **204**, or another stationary object. The primary embodiment is associated with free rotational movement of the driveshaft **208** and the load shaft **214**. In a second embodiment, the chamber **206** substantially rotates in unison with the driveshaft **208** (e.g., the chamber **206** is not fixed to a stationary object). In a third embodiment, the chamber **206** substantially rotates in unison with the load shaft **214** (e.g., the chamber **206** is not fixed to a stationary object).

[0072] The power coupler **200** further includes a snap ring **218** and a washer **220** combined to hold a first end of the driveshaft **208** in a fixed position with regards to the chamber **206**. Snap rings **222**

hold the drive turbine **210** to the other end of the driveshaft **208**. The power coupler **200** further includes a needle bearing **224** to facilitate rotation of the driveshaft **208** within the chamber **206**. The power coupler **200** further includes a seal **226** and a thrust bearing **228** to hold the drive turbine **210** in the fixed position with regards to the chamber **206**.

[0073] The drive turbine **210** includes an O-ring **230** and a where ring **232** to facilitate sealing of the drive turbine **210** within the chamber **206**. A key **234** facilitates mounting of the drive turbine **210** to the driveshaft **208**. In an embodiment, the driveshaft **208** and the drive turbine **210** are separate components. In another embodiment, the driveshaft **208** and the drive turbine **210** are manufactured as a common component. The drive turbine **210** is discussed in further detail with reference to FIG. 5B.

[0074] Within the chamber **206** and between the drive turbine **210** and the load turbine **212** is a shear thickening fluid (STF) **236**. Load turbine **212** includes O-rings **242** and a wear ring **244** to seal the load turbine **212** within the chamber **206**. The load shaft **214** extends from the load turbine **212** through the cartridge seal **216** and is held in place within the chamber by a washer **254** and snap rings **260** and **258**. In an embodiment, the load turbine **212** and the load shaft **214** are manufactured as a common component. In another embodiment, the load turbine **212** and the load shaft **214** are separate components. The cartridge seal **216** is held in place in an end of the chamber **206** in a fixed position by snap rings **250** and is sealed within the chamber **206** by o-rings **252**. The load shaft **214** rotates within the cartridge seal **216** facilitated by needle bearing **246**. The needle bearing **246** is secured within the cartridge seal **216** by a cap screw **248**.

[0075] The STF **236** is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates, wherein the second range of shear rates are greater than the first range of shear rates as discussed with reference to FIG. 1B. The STF **236** includes a plurality of nanoparticles that includes one or more of an oxide, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, SiO₂, polystyrene, polymethylmethacrylate, or a mixture of any of the above. The STF **236** further includes a solution to suspend the nanoparticles where the solution includes one or more of ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone, or a mixture of any of the above.

[0076] The chamber **206** is configured to contain a portion of the STF **236**, where the chamber includes a cylindrical interior channel. The cylindrical interior channel includes a drive side section and a load side section.

[0077] The drive shaft **208** is housed at least partially radially within the drive side section and protruding outward from a drive side section end of the chamber for coupling to the rotary power device **202**. The load shaft **214** is housed at least partially radially within the load side section and protruding outward from a load side section end of the chamber for coupling to the load device **204**.

[0078] The drive turbine **210** is housed at least partially radially within the drive side section and coupled to the drive shaft **208**. The drive turbine **210** is configured to exert pressure against the shear thickening fluid in response to rotary movement of the drive shaft from a rotary force applied to the drive shaft from the rotary power device.

[0079] The load turbine **212** is housed at least partially radially within the load side section at a fixed operational distance from the drive turbine and coupled to the load shaft. The load turbine configured to apply a secondary rotary force to the drive shaft in response to the pressure exerted against the shear thickening fluid from the drive turbine. The fixed operational distance between the drive turbine and the load turbine enables both the first range of shear rates and the second range of shear rates.

[0080] The cartridge seal **216** guides the load shaft **214** into the chamber **206**. The cartridge seal facilitates containment of the STF **236** within the chamber **206**. The cartridge seal remains in a fixed position relative to the chamber **206** (e.g., at an open end to facilitate manufacturability). The

snap ring **258** serves as a retaining device to maintain the load shaft **214** in a fixed position within the cartridge seal **216** to establish the fixed operational distance between the drive turbine **210** and the load turbine **212**.

[0081] FIG. 5B is a diagram of an embodiment of the drive turbine **210** that includes an O-ring channel **274**, a wear ring channel **272**, and a rotary array of drive teeth **270** arranged in a gear pattern. The gear pattern includes one or more of spur, helical, spiral, bevel, worm drive, and herringbone.

[0082] In an embodiment, the drive teeth **270** include at least one of a variety of configurations. A first configuration includes a first tooth configured with at least one cylindrical tube **276** with substantially consistent diameter from one side of the first tooth to an opposite side of the first tooth.

[0083] A second configuration includes a second tooth configured with at least one conical shaped tube **276** with an increasing diameter from one side of the second tooth to an opposite side of the second tooth. A third configuration includes a third tooth configured with at least one conical shaped tube with a decreasing diameter from one side of the third tooth to the opposite side of the third tooth. This conical shaping of the diameter creates the first and second levels of shear forces of the STF when the drive turbine is rotating within the STF of the chamber.

[0084] A fourth configuration includes a fourth tooth configured with at least one venturi shaped tube **278** from one side of the fourth tooth to the opposite side of the fourth tooth. This Venturi shaping of the tube also creates the first and second levels of shear forces of the STF when the drive turbine is rotating within the STF of the chamber.

[0085] The arranged gear pattern of the rotary array of drive teeth is configured to provide the decreasing viscosity in response to the first range of shear rates of the STF in the chamber in response to a first range of rotary power from the rotary power device. The range gear pattern also provides the increasing viscosity in response to the second range of shear rates of the STF in the chamber in response to a second range of rotary power from the rotary power device. The second range of rotary power is greater than the first range of rotary power.

[0086] FIG. 5C is a diagram of an embodiment of the load turbine **212** that includes the load shaft **214**, a wear-ring channel **282**, o-ring channels **284**, and a rotary array of load teeth **280**.

[0087] The rotary array of load teeth **280** are arranged in a gear pattern. The gear pattern includes one or more of spur, helical, spiral, bevel, worm drive, and herringbone. The gear pattern of the load turbine **212** complements the gear pattern of the rotary array of drive teeth of the drive turbine **210** such that the pressure exerted against the shear thickening fluid from the rotary array of drive teeth causes the rotary array of load teeth to apply the secondary rotary force to the drive shaft **214**.

[0088] In an embodiment, the load teeth **280** include a variety of configurations. A first configuration includes a first tooth configured with at least one cylindrical tube with substantially consistent diameter from one side of the first tooth to an opposite side of the first tooth (e.g., like the tube **276** of the drive turbine **210**).

[0089] A second configuration includes a second tooth configured with at least one conical shaped tube with an increasing diameter from one side of the second tooth to an opposite side of the second tooth. A third configuration includes a third tooth configured with at least one conical shaped tube with a decreasing diameter from one side of the third tooth to the opposite side of the third tooth (e.g., like the conical shaped tubes **276** of the drive turbine **210**). A fourth configuration includes a fourth tooth configured with at least one venturi shaped tube from one side of the fourth tooth to the opposite side of the fourth tooth (e.g., like the venturi tube **278** of the drive turbine **210**).

[0090] In another embodiment, none of the drive teeth **270** and none of the load teeth **280** include the above variety of configurations. In yet another embodiment, the drive teeth **270** include one or more holes and none of the load teeth **280** include holes. In a still further embodiment, none of the drive teeth **270** include holes and one or more of the load teeth **280** include one or more holes.

[0091] The arranged gear pattern of the rotary array of load teeth is configured to provide a first range of rotary output power to the load shaft in response to the first range of shear rates of the STF in the chamber resulting from a first range of rotary power from the rotary power device that causes the decreasing viscosity. The arranged gear pattern of the rotary array of load teeth also provides a second range of rotary output power to the load shaft in response to the second range of shear rates of the STF in the chamber resulting from a second range of rotary power from the rotary power device that causes the increasing viscosity. The second range of rotary output power is greater than the first range of rotary output power.

[0092] FIGS. 5D-5E are graphs portraying embodiments of operation of the rotary power coupler **200**. FIG. 5D illustrates driveshaft revolutions per minute (RPMs) versus load shaft RPMs. As the driveshaft begins to rotate and establishes a first threshold level of RPMs, the STF experiences the first range of shear rates and caused by the drive turbine which starts to move the load turbine within the first range of viscosity and hence the load shaft begins to turn.

[0093] As the driveshaft RPMs increase, a second threshold of driveshaft RPMs is established where the STF experiences the second range of shear rates caused by the drive turbine which moves the load turbine with increased force as the second range of viscosity is established and hence the load shaft begins to turn even more rapidly. For similar patterns of teeth of the drive turbine and load turbine, and as the second range of viscosity is established, the load shaft RPMs are similar to the driveshaft RPMs (e.g., approaching a 1:1 drive ratio) when the distance between the turbines is very close. The efficiency of the drive ratio is lowered as the turbines are separated since less STF is in play to move the load turbine and the amount of STF experiencing the second range of viscosity is lowered.

[0094] FIG. 5E also illustrates driveshaft revolutions per minute (RPMs) versus load shaft RPMs for an embodiment where the gear patterns are different between the drive turbine and the load turbine. The dotted line illustrates an embodiment where the load turbine is more efficient than the drive turbine (e.g., more teeth, more aggressive tapering of the tubes through the teeth). As the driveshaft begins to rotate and establishes a first threshold level of RPMs, the STF experiences the first range of shear rates and caused by the drive turbine which starts to move the load turbine within the first range of viscosity and hence the load shaft begins to turn.

[0095] As the driveshaft RPMs increase, a second threshold of driveshaft RPMs is established where the STF experiences the second range of shear rates caused by the drive turbine which moves the load turbine with increased force as the second range of viscosity is established and hence the load shaft begins to turn even more rapidly. For similar patterns of teeth of the drive turbine and load turbine, and as the second range of viscosity is established, the load shaft RPMs are similar to the driveshaft RPMs (e.g., approaching a 1:1 drive ratio). When the gear pattern of the load turbine is more efficient than the gear pattern of the drive turbine, the STF moved by the drive turbine moves the load turbine with more RPMs (e.g., higher than 1:1) during the utilization of the second viscosity range of the STF as illustrated by the dotted line.

[0096] FIG. 6A is a cross-section diagram of an embodiment of another rotary power coupler **300**. The power coupler **300** includes all the components of the power coupler **200** with the exception of the snap rings **258** and **260**. The power coupler **300** substitutes a shaft collar **256** for the snap rings **258** and **260**.

[0097] Within the chamber **206** of the power coupler **300** and between the drive turbine **210** and a load turbine **312** is a shear thickening fluid (STF) **236**. Load turbine **312** includes O-rings **242** and a wear ring **244** to seal the load turbine **312** within the chamber **206**. The load shaft **214** extends from the load turbine **312** through the cartridge seal **216** and is held in place within the chamber by the washer **254** and the shaft collar **256**. The cartridge seal **216** is held in place in an end of the chamber **206** in a fixed position by snap rings **250** and is sealed within the chamber **206** by o-rings **252**. The load shaft **214** rotates within the cartridge seal **216** facilitated by needle bearing **246**. The needle bearing **246** is secured within the cartridge seal **216** by a cap screw **248**.

[0098] The STF **236** is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates, wherein the second range of shear rates are greater than the first range of shear rates as discussed with reference to FIG. **1B**. The STF **236** includes a plurality of nanoparticles that includes one or more of an oxide, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, SiO₂, polystyrene, polymethylmethacrylate, or a mixture of any of the above. The STF **236** further includes a solution to suspend the nanoparticles where the solution includes one or more of ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone, or a mixture of any of the above.

[0099] The chamber **206** is configured to contain a portion of the STF **236**, where the chamber includes a cylindrical interior channel. The cylindrical interior channel includes a drive side section and a load side section.

[0100] A drive shaft **308** is housed at least partially radially within the drive side section and protruding outward from a drive side section end of the chamber for coupling to the rotary power device **202**. The load shaft **214** is housed at least partially radially within the load side section and protruding outward from a load side section end of the chamber for coupling to the load device **204**.

[0101] The drive turbine **210** is housed at least partially radially within the drive side section and coupled to the drive shaft **308**. The drive turbine **210** is configured to exert pressure against the shear thickening fluid in response to rotary movement of the drive shaft from a rotary force applied to the drive shaft from the rotary power device.

[0102] The load turbine **312** is housed at least partially radially within the load side section at an adjustable operational distance from the drive turbine and coupled to the load shaft. The load turbine is configured to apply a secondary rotary force to the drive shaft in response to the pressure exerted against the shear thickening fluid from the drive turbine. The adjustable operational distance between the drive turbine and the load turbine enables both the first range of shear rates and the second range of shear rates.

[0103] The shaft collar **256** is configured to establish the adjustable operational distance from the drive turbine to the load turbine. For example, a set screw is loosened to enable sliding the load shaft **214** either way and tightened once a desired operational distance is established.

[0104] The cartridge seal **216** guides the load shaft **214** into the chamber **206**. The cartridge seal facilitates containment of the STF **236** within the chamber **206**. The cartridge seal remains in a fixed position relative to the chamber **206** (e.g., at an open end to facilitate manufacturability).

[0105] FIG. **6B** is a graph portraying an embodiment of operation of the rotary power coupler **300** with closer spacing of the turbines as set by the shaft collar **256** illustrating driveshaft revolutions per minute (RPMs) versus load shaft RPMs. As the driveshaft begins to rotate and establishes a first threshold level of RPMs, the STF experiences the first range of shear rates and caused by the drive turbine which starts to move the load turbine within the first range of viscosity and hence the load shaft begins to turn.

[0106] As the driveshaft RPMs increase, a second threshold of driveshaft RPMs is quickly established due to the close spacing where the STF experiences the second range of shear rates caused by the drive turbine which moves the load turbine with increased force as the second range of viscosity is established and hence the load shaft begins to turn even more rapidly. For similar patterns of teeth of the drive turbine and load turbine, and as the second range of viscosity is established, the load shaft RPMs are similar to the driveshaft RPMs (e.g., approaching a 1:1 drive ratio) when the distance between the turbines is very close.

[0107] FIG. **6C** is a cross-section diagram of another embodiment of the rotary power coupler **300** where the turbines are separated by a greater distance than as illustrated in FIG. **6A**. As such, lower shear forces are in play for the shear thickening fluid **236** between the drive turbine **210** and the load turbine **312**. The shaft collar **256** is field adjusted to establish the further distance between the

turbines.

[0108] FIG. 6D is a graph portraying another embodiment of operation of the rotary power coupler **300** with further spacing of the turbines as set by the shaft collar **256** illustrating driveshaft revolutions per minute (RPMs) versus load shaft RPMs. As the driveshaft begins to rotate and establishes a new first threshold level of RPMs, the STF experiences the first range of shear rates and caused by the drive turbine which starts to move the load turbine within the first range of viscosity and hence the load shaft begins to turn.

[0109] As the driveshaft RPMs increase, a new second threshold of driveshaft RPMs is finally established due to the further spacing where the STF begins to experience the second range of shear rates caused by the drive turbine which moves the load turbine with some increased force as the second range of viscosity is established and hence the load shaft begins to turn more rapidly. For similar patterns of teeth of the drive turbine and load turbine, and as the second range of viscosity is at least in part established, the load shaft RPMs approach a maximum percentage of the driveshaft RPMs (e.g., approaching something less than a 1:1 drive ratio) when the distance between the turbines is further away.

[0110] FIGS. 7A-7B are cross-section diagrams of an embodiment of another rotary power coupler **400** illustrating an example of controlling coupling of rotary power. The power coupler **400** includes the components of the power coupler **300** with the exception of a load actuator **456** replacing the shaft collar **256** and an actuator track **458** associated with the load shaft **214**. The power coupler **400** further includes the computing entity **20-1** of FIG. 1, the environment sensor module **14** of FIG. 2B, the sensor **116** of FIG. 1, and the emitter **114** of FIG. 1.

[0111] The load actuator **456** is configured to establish an adjustable operational distance from the drive turbine to the load turbine. For example, a motor of the load actuator **456** moves the actuator track **458** to push or pull the load shaft **214** within the chamber **206**.

[0112] The sensor **116** includes a set of fluid flow sensors to produce a fluid response of the STF **236**. The set of fluid flow sensors are positioned proximal to the chamber **206**. The set of fluid flow sensors includes a load turbine position sensor, a load actuator position sensor, and an actuator track position sensor.

[0113] The emitter **114** includes a set of fluid manipulation emitters positioned proximal to the chamber **206**. The set of fluid manipulation emitters provide a fluid activation (e.g., selecting a shear rate range of the STF) to at least one of the STF **236** and the load actuator **456** to enable selection of the first range of shear rates and the second range of shear rates.

[0114] The computing entity **20-1** includes the control module **30** in the chamber database **34** of FIG. 1. The environment sensor module **14** senses sensed factors **460** associated with the load device **204** and provides environmental sensor information **150** to the computing entity **20-1**. The sensed factors **460** includes RPMs associated with the load device **204**, any movement of an object associated with the load device **204**, and anything associated with another object indirectly associated with the load device **204** within a common environment.

[0115] FIG. 7A illustrates steps of an example method of operation of the power coupler **400** where a first step includes the computing entity **20-1** activating the power coupler **400** using a baseline STF activation **470**. An STF activation includes creating an environment within the chamber **206** for the STF to experience a desired first or second range of shear rates and hence the first or second range of viscosities of the STF to provide a desired level of power coupling from the rotary power device **202** to the load device **204**. A baseline STF activation includes an initial set of STF activations that are expected to produce the desired level of power coupling.

[0116] The set of baseline STF activation includes providing a default level of power in a power control **466** signal to the rotary power device **202**, outputting a default fluid activation signal to the emitter **114** to modify the STF for an initial shear force versus viscosity curve, and moving the load shaft **214** via the load actuator **456** to a starting position for a default separation of the turbines that is expected to experience a desired level of shear forces of the STF between the turbines. For

instance, the control module **30** outputs the power control **466** to the rotary power device **202**, outputs the fluid activation **234** to the emitter **114**, and outputs actuator control **462** to the load actuator **456** to achieve the desired turbine spacing.

[0117] Having activated the power coupler **400** using the baseline STF activation, a second step of the example method of operation includes the computing entity **20-1** determining a coupler status **472** of the power coupler **400**. The status of the power coupler **400** includes one or more of a power conversion factor between the rotary power device **202** and the load device **204**, RPMs of the rotary power device **202** verses RPMs of the load device **204**, the STF fluid response (e.g., which range of shear rates at present), position and utilization information associated with the load device **204**, and information with regards to the sensed factors **460**. The determining of the status includes a variety of alternatives.

[0118] In a first alternative example, the control module **30** interprets fluid response **232** from the sensor **116** to determine status of shear rates of the STF. As another alternative example, the control module **30** interprets load information **464** from the load device **204** to determine the position and utilization information associated with the load device **204**. As yet another alternative example, the control module **30** interprets environment sensor information **150** from the environment sensor module **14** to determine results from the sensed factors **460** with regards to movement of any object associated with the load device **204**.

[0119] FIG. 7B further illustrates the example method of operation where in a third step the computing entity **20-1** determines an updated STF activation **474** based on the status of the power coupler **400**. The determining includes identifying a desired status of the power coupler and determining changes to affect the power coupler to move from the current status to the desired status. The identifying the desired status includes at least one of performing a lookup, interpreting a received message, and determining the desired status. For example, the control module **30** retrieves the desired status from the chamber database **34** for load device **204**.

[0120] The determining the changes to affect the power coupler includes a variety of alternatives. A first alternative includes increasing the STF viscosity when delivered power is less than desired (e.g., an object associated with the load device **204** is not moving fast enough). In an instance, the control module **30** determines a new fluid activation **234** to modify the STF for a higher viscosity. As another instance, the control module **30** determines to reduce the gap between the turbines to increase the viscosity. As yet another example, the control module **30** determines to increase power to the rotary power device **202** to raise the delivered power.

[0121] A second alternative includes decreasing the STF viscosity when the delivered power is greater than desired (e.g., the object associated with the load device is moving too fast). In an instance, the control module **30** determines the new fluid activation **234** to modify the STF for a lower viscosity. As another instance, the control module **30** determines to increase the gap between the turbines to decrease the viscosity. As yet another example, the control module **30** determines to lower the power to the rotary power device **202** to lower the delivered power.

[0122] Having determined the updated STF activation, a fourth step of the example method of operation includes the computing entity **20-1** applying the updated STF activation to the power coupler **400** to facilitate the transferring of the rotary power from the rotary power device to the load device. For example, the control module **30** outputs an updated actuator control **462** to the load actuator **456** in accordance with the updated STF activation to facilitate an updated separation of the turbines (e.g., closer for more viscosity to transfer more power for speeding up or even hard braking, further away for less viscosity to slow down or to coast). As another example, the control module **30** outputs an updated fluid activation **234** to the emitter **114** to modify the viscosity of the STF in accordance with the updated STF activation (e.g., more viscosity to transfer more power for speeding up or hard braking, less viscosity to slow down or to coast). As yet another example, the control module **30** outputs an updated power control **466** to the rotary power device **202** to either increase or decrease the power in accordance with the updated STF activation.

[0123] FIG. 7C is a timing diagram set portraying an embodiment of operation of rotary power coupler **400**. With the driveshaft running turning at a reference 1.0 and the gap between the turbines at a baseline separation, the STF between the turbines engages such that the load shaft turns at a reference rate of 0.5.

[0124] When the desired power transfer is to spin the load shaft at a higher speed, the gap between the turbines is lowered to speed up the load turbine such that the load shaft turns at a reference rate of 0.9 (e.g., almost 1:1 ratio). Later, the gap between the turbines is widened such that the load turbine and load shaft spin slower at a reference rate of 0.3. Next the power to the rotary power device is turned off and the distance between the turbines is closed up to a very close gap such that the load shaft spends down very quickly in unison with the drive shaft to provide braking.

[0125] The method described above in conjunction with a processing module of any computing entity of the mechanical and computing system of FIG. 7A can alternatively be performed by other modules of the system of FIG. 7A or by other devices. In addition, at least one memory section that is non-transitory (e.g., a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element, a sixth memory element, etc.) that stores operational instructions can, when executed by one or more processing modules of the one or more computing entities of the computing system, cause one or more computing devices of the mechanical and computing system of FIG. 7A to perform any or all of the method steps described above.

[0126] FIG. 8A is a cross-section diagram of an embodiment of a rotary power shunt **500** that includes all the components of the power coupler **200** along with a lock **502** and pins **506**. The power shunt **500** shunts rotational power from a load device **504** providing a braking action as desired. The load device **504** includes any type of device powered by and/or using rotary motion such as a vehicle, a wheel, an elevator, a conveyor system, an automatic door, etc.

[0127] The STF **236** is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates, wherein the second range of shear rates are greater than the first range of shear rates as discussed with reference to FIG. 1B. The STF **236** includes a plurality of nanoparticles that includes one or more of an oxide, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, SiO₂, polystyrene, polymethylmethacrylate, or a mixture of any of the above. The STF **236** further includes a solution to suspend the nanoparticles where the solution includes one or more of ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone, or a mixture of any of the above.

[0128] The chamber **206** is configured to contain a portion of the STF **236**, where the chamber includes a cylindrical interior channel. The cylindrical interior channel includes a drive side section and a load side section.

[0129] In an embodiment, the drive shaft **208** is housed at least partially radially within the drive side section and protruding outward from a drive side section end of the chamber for coupling to lock **502** configured to prevent rotation of the driveshaft **208**. The load shaft **214** is housed at least partially radially within the load side section and protruding outward from a load side section end of the chamber for coupling to the load device **204**.

[0130] In the embodiment, the drive turbine **210** is housed at least partially radially within the drive side section and coupled to the drive shaft **208**. The drive turbine **210** is configured to exert resistive pressure against the shear thickening fluid in response to rotary movement of the load shaft from a rotary force applied to the load shaft from the load device.

[0131] In another embodiment, a fixed position of the drive turbine **210** with respect to the chamber **206** is provided by one or more of configuring the drive turbine **210** and the chamber **206** as substantially one component and configuring the drive turbine **210**, the chamber **206**, and the driveshaft **208** as substantially one component. In yet another embodiment, the driveshaft **208** is

coupled to a stationary external object to provide the fixed position of the drive turbine **210**. The stationary external object includes at least one of a lock, the rotary power device **202** locked in a stop position, and a servo motor that is held in a stopped position.

[0132] The load turbine **212** is housed at least partially radially within the load side section at a fixed operational distance from the drive turbine and coupled to the load shaft. The load turbine is configured to apply at least some of the rotary power from the load device via the load shaft to the STF. The STF, in response to the pressure exerted against the STF from the load turbine, exerts pressure on the drive turbine. The fixed operational distance between the drive turbine and the load turbine enables both the first range of shear rates and the second range of shear rates.

[0133] The cartridge seal **216** guides the load shaft **214** into the chamber **206**. The cartridge seal facilitates containment of the STF **236** within the chamber **206**. The cartridge seal remains in a fixed position relative to the chamber **206** (e.g., at an open end to facilitate manufacturability). The snap ring **258** serves as a retaining device to maintain the load shaft **214** in a fixed position within the cartridge seal **216** to establish the fixed operational distance between the drive turbine **210** and the load turbine **212**.

[0134] The drive turbine **210** includes the rotary array of drive teeth arranged in the gear pattern. The arranged gear pattern of the rotary array of drive teeth of the power shunt **500** is configured to provide a first range of rotary output power to the drive shaft in response to the first range of shear rates of the STF in the chamber resulting from a first range of rotary power from the load device that causes the decreasing viscosity.

[0135] The arranged gear pattern of the rotary array of drive teeth of the power shunt **500** is further configured to provide a second range of rotary output power to the drive shaft in response to the second range of shear rates of the STF in the chamber resulting from a second range of rotary power from the load device that causes the increasing viscosity. The second range of rotary output power is greater than the first range of rotary output power. The operation of the power shunt **500** to provide braking power is further discussed with reference to FIG. **8B**.

[0136] The load turbine **212** includes the rotary array of load teeth arranged in the gear pattern. The arranged gear pattern of the rotary array of load teeth of the power shunt **500** is configured to provide the decreasing viscosity in response to the first range of shear rates of the STF in the chamber in response to a first range of rotary power from the load device. The arranged gear pattern of the rotary array of load teeth of the power shunt **500** is further configured to provide the increasing viscosity in response to the second range of shear rates of the STF in the chamber in response to a second range of rotary power from the load device. The operation of the power shunt **500** to provide braking power is further discussed with reference to FIG. **8B**.

[0137] FIG. **8B** is a graph portraying an embodiment of operation of a rotary power shunt illustrating turbine engagement versus braking force. The turbine engagement indicates proximity of the turbines, where a maximum level of turbine engagement means that the turbines are right next to each other with substantially no gap. No engagement means that the turbines are separated so much that there is virtually no transfer of power via the STF between the turbines.

[0138] In an example of operation where a load speed of the load device **504** is a nominal level, STF engagement at a low and below threshold effect level starts as the turbine engagement is increased. Minimal braking force is achieved. Further braking force rapidly increases once the STF reaches the second threshold and the second viscosity range where the load turbine transfers more power to the drive turbine for shunting. At the closest turbine engagement, a massive STF engagement occurs where having already achieved the second range of viscosity, simply more volume of the STF is compressed and available to provide drive from the load turbine to the drive turbine to provide more braking force.

[0139] When the load speed is slower than the nominal load speed, the turbine engagement must be higher to achieve similar braking forces as illustrated by the gray line to the right. When the load speed is faster than the nominal load speed, the turbine engagement can be lowered to achieve

similar braking forces as illustrated by the gray line to the left.

[0140] The power shunt **500** may be utilized as a governor of sorts by selecting a turbine engagement level to provide constant braking power to the load device **504** whenever it is above the threshold of the STF as it moves from the first range of viscosities to the second range of viscosities.

[0141] FIG. **9A** is a cross-section diagram of another embodiment of a rotary power shunt **600** that includes all the components of the power coupler **300** along with the lock **502** and the pins **506**. The power shunt **600** shunts rotational power from a load device **504** providing a braking action as desired. The load device **504** includes any type of device powered by and/or using rotary motion such as a vehicle, a wheel, an elevator, a conveyor system, an automatic door, etc.

[0142] The STF **236** is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates, wherein the second range of shear rates are greater than the first range of shear rates as discussed with reference to FIG. **1B**. The STF **236** includes a plurality of nanoparticles that includes one or more of an oxide, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, SiO₂, polystyrene, polymethylmethacrylate, or a mixture of any of the above. The STF **236** further includes a solution to suspend the nanoparticles where the solution includes one or more of ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone, or a mixture of any of the above.

[0143] The chamber **206** is configured to contain a portion of the STF **236**, where the chamber includes a cylindrical interior channel. The cylindrical interior channel includes a drive side section and a load side section.

[0144] The drive shaft **208** is housed at least partially radially within the drive side section and protruding outward from a drive side section end of the chamber for coupling to lock **502** configured to prevent rotation of the driveshaft **208**. The load shaft **214** is housed at least partially radially within the load side section and protruding outward from a load side section end of the chamber for coupling to the load device **204**.

[0145] The drive turbine **210** is housed at least partially radially within the drive side section and coupled to the drive shaft **208**. The drive turbine **210** is configured to exert resistive pressure against the shear thickening fluid in response to rotary movement of the load shaft from a rotary force applied to the load shaft from the load device.

[0146] The load turbine **212** is housed at least partially radially within the load side section at an adjustable operational distance from the drive turbine and coupled to the load shaft. The load turbine is configured to apply at least some of the rotary power from the load device via the load shaft to the STF. The STF, in response to the pressure exerted against the STF from the load turbine, exerts pressure on the drive turbine. The adjustable operational distance between the drive turbine and the load turbine enables both the first range of shear rates and the second range of shear rates.

[0147] The shaft collar **256** is configured to establish the adjustable operational distance from the drive turbine to the load turbine is a primary mechanism to establish turbine engagement and utilization of desired STF viscosity ranges. For example, a set screw can be field adjusted to enable moving and then establishing the gap between the turbines at a desired level.

[0148] The cartridge seal **216** guides the load shaft **214** into the chamber **206**. The cartridge seal facilitates containment of the STF **236** within the chamber **206**. The cartridge seal remains in a fixed position relative to the chamber **206** (e.g., at an open end to facilitate manufacturability).

[0149] The drive turbine **210** includes the rotary array of drive teeth arranged in the gear pattern. The arranged gear pattern of the rotary array of drive teeth of the power shunt **600** is configured to provide a first range of rotary output power to the drive shaft in response to the first range of shear rates of the STF in the chamber resulting from a first range of rotary power from the load device that causes the decreasing viscosity.

[0150] The arranged gear pattern of the rotary array of drive teeth of the power shunt **600** is further configured to provide a second range of rotary output power to the drive shaft in response to the second range of shear rates of the STF in the chamber resulting from a second range of rotary power from the load device that causes the increasing viscosity. The second range of rotary output power is greater than the first range of rotary output power. The operation of the power shunt **600** to provide braking power is further discussed with reference to FIG. **9B**.

[0151] The load turbine **212** includes the rotary array of load teeth arranged in the gear pattern. The arranged gear pattern of the rotary array of load teeth of the power shunt **600** is configured to provide the decreasing viscosity in response to the first range of shear rates of the STF in the chamber in response to a first range of rotary power from the load device. The arranged gear pattern of the rotary array of load teeth of the power shunt **600** is further configured to provide the increasing viscosity in response to the second range of shear rates of the STF in the chamber in response to a second range of rotary power from the load device. The operation of the power shunt **600** to provide braking power is further discussed with reference to FIG. **9B**.

[0152] FIG. **9B** is a graph portraying an embodiment of operation of a rotary power shunt illustrating turbine engagement versus braking force. The turbine engagement indicates proximity of the turbines, where a maximum level of turbine engagement means that the turbines are right next to each other with substantially no gap. No engagement means that the turbines are separated so much from a setting of the shaft collar **256** that there is virtually no transfer of power via the STF between the turbines.

[0153] In an example of operation where a load speed of the load device **504** is a nominal level, STF engagement at a low and below threshold effect level starts as the turbine engagement is increased utilizing shaft collar **256**. Minimal braking force is achieved. Further braking force rapidly increases once the STF reaches the second threshold and the second viscosity range where the load turbine transfers more power to the drive turbine for shunting. At the closest turbine engagement set by shaft collar **256**, a massive STF engagement occurs where having already achieved the second range of viscosity, simply more volume of the STF is compressed and available to provide drive from the load turbine to the drive turbine to provide more braking force.

[0154] When the load speed is slower than the nominal load speed, the turbine engagement must be higher to achieve similar braking forces as illustrated by the gray line to the right. When the load speed is faster than the nominal load speed, the turbine engagement can be lowered to achieve similar braking forces as illustrated by the gray line to the left.

[0155] The power shunt **600** may be utilized as a governor of sorts by selecting a turbine engagement level to provide constant braking power to the load device **504** whenever it is above the threshold of the STF as it moves from the first range of viscosities to the second range of viscosities. The governor can be adjusted by moving the shaft collar **256** to achieve a desired level of engagement of the governor.

[0156] FIGS. **10A-10B** are cross-section diagrams of an embodiment of another rotary power shunt **700** illustrating an example of controlling shunting of rotary power. The power shunt **700** includes all the components of the power coupler **400** along with the lock **502** and the pins **506**. The power shunt **700** shunts rotational power from a load device **504** providing an automation of braking action as desired. The load device **504** includes any type of device powered by and/or using rotary motion such as a vehicle, a wheel, an elevator, a conveyor system, an automatic door, etc.

[0157] The STF **236** is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates, wherein the second range of shear rates are greater than the first range of shear rates as discussed with reference to FIG. **1B**. The STF **236** includes a plurality of nanoparticles that includes one or more of an oxide, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, SiO₂, polystyrene, polymethylmethacrylate, or a mixture of any of the above. The STF **236** further includes a solution to suspend the nanoparticles where the solution includes one or more of

ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone, or a mixture of any of the above.

[0158] The chamber **206** is configured to contain a portion of the STF **236**, where the chamber includes a cylindrical interior channel. The cylindrical interior channel includes a drive side section and a load side section.

[0159] The drive shaft **208** is housed at least partially radially within the drive side section and protruding outward from a drive side section end of the chamber for coupling to lock **502** configured to prevent rotation of the driveshaft **208**. The load shaft **214** is housed at least partially radially within the load side section and protruding outward from a load side section end of the chamber for coupling to the load device **204**.

[0160] The drive turbine **210** is housed at least partially radially within the drive side section and coupled to the drive shaft **208**. The drive turbine **210** is configured to exert resistive pressure against the shear thickening fluid in response to rotary movement of the load shaft from a rotary force applied to the load shaft from the load device.

[0161] The load turbine **212** is housed at least partially radially within the load side section at an adjustable operational distance from the drive turbine and coupled to the load shaft. The load turbine is configured to apply at least some of the rotary power from the load device via the load shaft to the STF. The STF, in response to the pressure exerted against the STF from the load turbine, exerts pressure on the drive turbine. The adjustable operational distance between the drive turbine and the load turbine enables both the first range of shear rates and the second range of shear rates.

[0162] The load actuator **456** is configured to establish the adjustable operational distance from the drive turbine to the load turbine. For example, a motor of the load actuator **456** moves the actuator track **458** to push or pull the load shaft **214** within the chamber **206**.

[0163] The cartridge seal **216** guides the load shaft **214** into the chamber **206**. The cartridge seal facilitates containment of the STF **236** within the chamber **206**. The cartridge seal remains in a fixed position relative to the chamber **206** (e.g., at an open end to facilitate manufacturability).

[0164] FIG. **10A** illustrates an example method of operation of the shunting of the power where a first step includes the computing entity **20-1** determining a shunt status **772** of the power shunt **700**. The determining of the status of the power shunt includes one or more of interpreting the fluid response **232** from the sensor **116**, interpreting load information **464** from the load device **204**, and interpreting environment sensor information **150** from the environment sensor module **14** with regards to sensed factors **460**. For example, the control module **30** determines a speed of an object associated with the load device **204** from the load information **464**. The determining of the status further includes recovering the baseline STF activation **470** from the chamber database **34**.

[0165] FIG. **10B** illustrates a second step of the example method of operation of the shunting of the power where the computing entity **20-1** determines an updated STF activation or **74** based on the power shunt status **772**. For example, the control module **30** determines to increase the STF viscosity when braking power is less than desired or when an object associated with the load device **204** is moving too fast. As another example, the control module **30** determines to decrease viscosity when the braking power is greater than desired or when the object associated with the load device is moving too slow. The updated STF activation includes one or more of an updated fluid activation **234** to directly adjust the viscosity of the STF by way of the emitter **114** and an updated actuator control **462** to adjust the separation of the turbines to change the viscosity range.

[0166] Having determined the updated STF activation **474**, a third step of the example method of operation includes the computing entity **20-1** activating the power shunt **700** using the updated STF activation **474**. For example, the control module **30** outputs the fluid activation **234** to the emitter **114** for an updated shear force versus viscosity to add or reduce braking power. As another example, the control module **30** outputs the actuator control **462** to power the load actuator **456** to move the load shaft **214** for an adjusted separation of the turbines. For instance, the turbines are

moved closer to increase the braking power and are moved further apart to lower the braking power to coast or speed up on its own.

[0167] FIG. **10C** is a timing diagram set portraying an embodiment of operation of the rotary power shunt **700** where a baseline separation of the turbines provides a baseline load shaft speed. When a closer gap between the turbines is produced the load shaft revolutions is reduced. When the gap between the turbines is increased beyond the baseline the load shaft revolutions speeds up faster than the baseline speed. When the gap between the turbines is significantly reduced the load shaft experiences hard braking.

[0168] The method described above in conjunction with a processing module of any computing entity of the mechanical and computing system of FIG. **10A** can alternatively be performed by other modules of the system of FIG. **10A** or by other devices. In addition, at least one memory section that is non-transitory (e.g., a non-transitory computer readable storage medium, a non-transitory computer readable memory organized into a first memory element, a second memory element, a third memory element, a fourth element section, a fifth memory element, a sixth memory element, etc.) that stores operational instructions can, when executed by one or more processing modules of the one or more computing entities of the computing system, cause one or more computing devices of the mechanical and computing system of FIG. **10A** to perform any or all of the method steps described above.

[0169] It is noted that terminologies as may be used herein such as bit stream, stream, signal sequence, etc. (or their equivalents) have been used interchangeably to describe digital information whose content corresponds to any of a number of desired types (e.g., data, video, speech, text, graphics, audio, etc. any of which may generally be referred to as 'data').

[0170] As may be used herein, the terms "substantially" and "approximately" provides an industry-accepted tolerance for its corresponding term and/or relativity between items. For some industries, an industry-accepted tolerance is less than one percent and, for other industries, the industry-accepted tolerance is 10 percent or more. Other examples of industry-accepted tolerance range from less than one percent to fifty percent. Industry-accepted tolerances correspond to, but are not limited to, component values, integrated circuit process variations, temperature variations, rise and fall times, thermal noise, dimensions, signaling errors, dropped packets, temperatures, pressures, material compositions, and/or performance metrics. Within an industry, tolerance variances of accepted tolerances may be more or less than a percentage level (e.g., dimension tolerance of less than $\pm 1\%$). Some relativity between items may range from a difference of less than a percentage level to a few percent. Other relativity between items may range from a difference of a few percent to magnitude of differences.

[0171] As may also be used herein, the term(s) "configured to", "operably coupled to", "coupled to", and/or "coupling" includes direct coupling between items and/or indirect coupling between items via an intervening item (e.g., an item includes, but is not limited to, a component, an element, a circuit, and/or a module) where, for an example of indirect coupling, the intervening item does not modify the information of a signal but may adjust its current level, voltage level, and/or power level. As may further be used herein, inferred coupling (i.e., where one element is coupled to another element by inference) includes direct and indirect coupling between two items in the same manner as "coupled to".

[0172] As may even further be used herein, the term "configured to", "operable to", "coupled to", or "operably coupled to" indicates that an item includes one or more of power connections, input(s), output(s), etc., to perform, when activated, one or more its corresponding functions and may further include inferred coupling to one or more other items. As may still further be used herein, the term "associated with", includes direct and/or indirect coupling of separate items and/or one item being embedded within another item.

[0173] As may be used herein, the term "compares favorably", indicates that a comparison between two or more items, signals, etc., provides a desired relationship. For example, when the desired

relationship is that signal 1 has a greater magnitude than signal 2, a favorable comparison may be achieved when the magnitude of signal 1 is greater than that of signal 2 or when the magnitude of signal 2 is less than that of signal 1. As may be used herein, the term “compares unfavorably”, indicates that a comparison between two or more items, signals, etc., fails to provide the desired relationship.

[0174] As may be used herein, one or more claims may include, in a specific form of this generic form, the phrase “at least one of a, b, and c” or of this generic form “at least one of a, b, or c”, with more or less elements than “a”, “b”, and “c”. In either phrasing, the phrases are to be interpreted identically. In particular, “at least one of a, b, and c” is equivalent to “at least one of a, b, or c” and shall mean a, b, and/or c. As an example, it means: “a” only, “b” only, “c” only, “a” and “b”, “a” and “c”, “b” and “c”, and/or “a”, “b”, and “c”.

[0175] As may also be used herein, the terms “processing module”, “processing circuit”, “processor”, “processing circuitry”, and/or “processing unit” may be a single processing device or a plurality of processing devices. Such a processing device may be a microprocessor, micro-controller, digital signal processor, microcomputer, central processing unit, field programmable gate array, programmable logic device, state machine, logic circuitry, analog circuitry, digital circuitry, and/or any device that manipulates signals (analog and/or digital) based on hard coding of the circuitry and/or operational instructions. The processing module, module, processing circuit, processing circuitry, and/or processing unit may be, or further include, memory and/or an integrated memory element, which may be a single memory device, a plurality of memory devices, and/or embedded circuitry of another processing module, module, processing circuit, processing circuitry, and/or processing unit. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, and/or any device that stores digital information. Note that if the processing module, module, processing circuit, processing circuitry, and/or processing unit includes more than one processing device, the processing devices may be centrally located (e.g., directly coupled together via a wired and/or wireless bus structure) or may be distributedly located (e.g., cloud computing via indirect coupling via a local area network and/or a wide area network).

Further note that if the processing module, module, processing circuit, processing circuitry and/or processing unit implements one or more of its functions via a state machine, analog circuitry, digital circuitry, and/or logic circuitry, the memory and/or memory element storing the corresponding operational instructions may be embedded within, or external to, the circuitry comprising the state machine, analog circuitry, digital circuitry, and/or logic circuitry. Still further note that, the memory element may store, and the processing module, module, processing circuit, processing circuitry and/or processing unit executes, hard coded and/or operational instructions corresponding to at least some of the steps and/or functions illustrated in one or more of the Figures. Such a memory device or memory element can be included in an article of manufacture.

[0176] One or more embodiments have been described above with the aid of method steps illustrating the performance of specified functions and relationships thereof. The boundaries and sequence of these functional building blocks and method steps have been arbitrarily defined herein for convenience of description. Alternate boundaries and sequences can be defined so long as the specified functions and relationships are appropriately performed. Any such alternate boundaries or sequences are thus within the scope and spirit of the claims. Further, the boundaries of these functional building blocks have been arbitrarily defined for convenience of description. Alternate boundaries could be defined as long as the certain significant functions are appropriately performed. Similarly, flow diagram blocks may also have been arbitrarily defined herein to illustrate certain significant functionality.

[0177] To the extent used, the flow diagram block boundaries and sequence could have been defined otherwise and still perform the certain significant functionality. Such alternate definitions of both functional building blocks and flow diagram blocks and sequences are thus within the

scope and spirit of the claims. One of average skill in the art will also recognize that the functional building blocks, and other illustrative blocks, modules, and components herein, can be implemented as illustrated or by discrete components, application specific integrated circuits, processors executing appropriate software and the like or any combination thereof.

[0178] In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with one or more other routines. In addition, a flow diagram may include an “end” and/or “continue” indication. The “end” and/or “continue” indications reflect that the steps presented can end as described and shown or optionally be incorporated in or otherwise used in conjunction with one or more other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

[0179] The one or more embodiments are used herein to illustrate one or more aspects, one or more features, one or more concepts, and/or one or more examples. A physical embodiment of an apparatus, an article of manufacture, a machine, and/or of a process may include one or more of the aspects, features, concepts, examples, etc. described with reference to one or more of the embodiments discussed herein. Further, from figure to figure, the embodiments may incorporate the same or similarly named functions, steps, modules, etc. that may use the same or different reference numbers and, as such, the functions, steps, modules, etc. may be the same or similar functions, steps, modules, etc. or different ones.

[0180] Unless specifically stated to the contra, signals to, from, and/or between elements in a figure of any of the figures presented herein may be analog or digital, continuous time or discrete time, and single-ended or differential. For instance, if a signal path is shown as a single-ended path, it also represents a differential signal path. Similarly, if a signal path is shown as a differential path, it also represents a single-ended signal path. While one or more particular architectures are described herein, other architectures can likewise be implemented that use one or more data buses not expressly shown, direct connectivity between elements, and/or indirect coupling between other elements as recognized by one of average skill in the art.

[0181] The term “module” is used in the description of one or more of the embodiments. A module implements one or more functions via a device such as a processor or other processing device or other hardware that may include or operate in association with a memory that stores operational instructions. A module may operate independently and/or in conjunction with software and/or firmware. As also used herein, a module may contain one or more sub-modules, each of which may be one or more modules.

[0182] As may further be used herein, a computer readable memory includes one or more memory elements. A memory element may be a separate memory device, multiple memory devices, or a set of memory locations within a memory device. Such a memory device may be a read-only memory, random access memory, volatile memory, non-volatile memory, static memory, dynamic memory, flash memory, cache memory, a quantum register or other quantum memory and/or any other device that stores data in a non-transitory manner. Furthermore, the memory device may be in a form of a solid-state memory, a hard drive memory or other disk storage, cloud memory, thumb drive, server memory, computing device memory, and/or other non-transitory medium for storing data. The storage of data includes temporary storage (i.e., data is lost when power is removed from the memory element) and/or persistent storage (i.e., data is retained when power is removed from the memory element). As used herein, a transitory medium shall mean one or more of: (a) a wired or wireless medium for the transportation of data as a signal from one computing device to another computing device for temporary storage or persistent storage; (b) a wired or wireless medium for

the transportation of data as a signal within a computing device from one element of the computing device to another element of the computing device for temporary storage or persistent storage; (c) a wired or wireless medium for the transportation of data as a signal from one computing device to another computing device for processing the data by the other computing device; and (d) a wired or wireless medium for the transportation of data as a signal within a computing device from one element of the computing device to another element of the computing device for processing the data by the other element of the computing device. As may be used herein, a non-transitory computer readable memory is substantially equivalent to a computer readable memory. A non-transitory computer readable memory can also be referred to as a non-transitory computer readable storage medium.

[0183] While particular combinations of various functions and features of the one or more embodiments have been expressly described herein, other combinations of these features and functions are likewise possible. The present disclosure is not limited by the particular examples disclosed herein and expressly incorporates these other combinations.

Claims

1. A power shunt for shunting rotary power from a load device, the power shunt comprising: a shear thickening fluid (STF), wherein the STF is configured to have a decreasing viscosity in response to a first range of shear rates and an increasing viscosity in response to a second range of shear rates, wherein the second range of shear rates are greater than the first range of shear rates; a chamber, the chamber configured to contain a portion of the STF, wherein the chamber includes a cylindrical interior channel, wherein the cylindrical interior channel includes a drive side section and a load side section; a drive shaft, the drive shaft housed at least partially radially within the drive side section and protruding outward from a drive side section end of the chamber for coupling to a lock configured to prevent rotation of the drive shaft; a load shaft, the load shaft housed at least partially radially within the load side section and protruding outward from a load side section end of the chamber through a shaft collar for coupling to the load device; a drive turbine, the drive turbine housed at least partially radially within the drive side section and coupled to the drive shaft, the drive turbine configured to exert resistive pressure against the shear thickening fluid in response to rotary movement of the load shaft from a rotary force applied to the load shaft from the load device, wherein the drive turbine includes a rotary array of drive teeth arranged in a gear pattern of the drive teeth; and a load turbine, the load turbine housed at least partially radially within the load side section at an adjustable operational distance from the drive turbine and coupled to the load shaft, the load turbine configured to apply at least some of the rotary power from the load device via the load shaft to the STF, wherein the STF, in response to pressure exerted against the shear thickening fluid from the load turbine, exerts pressure on the drive turbine, wherein the shaft collar is configured to establish the adjustable operational distance from the drive turbine to the load turbine, wherein the adjustable operational distance between the drive turbine and the load turbine enables both the first range of shear rates and the second range of shear rates, wherein the load turbine includes a rotary array of load teeth arranged in a gear pattern of the load teeth, wherein the gear pattern of the load teeth complements the gear pattern of the drive teeth such that pressure exerted against the shear thickening fluid from the rotary array of load teeth causes the rotary array of drive teeth to apply a secondary rotary force to the drive shaft such that revolutions per unit of time of the load shaft are greater than revolutions per the unit of time of the drive shaft when the adjustable operational distance from the drive turbine to the load turbine enables the second range of shear rates.

2. The power shunt of claim 1 further comprises: a cartridge seal to guide the load shaft into the chamber, wherein the cartridge seal facilitates containment of the STF within the chamber, wherein the cartridge seal remains in a fixed position relative to the chamber.

3. The power shunt of claim 1, wherein the STF comprises: a plurality of nanoparticles, wherein the plurality of nanoparticles includes one or more of an oxide, calcium carbonate, synthetically occurring minerals, naturally occurring minerals, polymers, SiO₂, polystyrene, polymethylmethacrylate, or a mixture thereof.
 4. The power shunt of claim 1, wherein the STF further comprises: one or more of ethylene glycol, polyethylene glycol, ethanol, silicon oils, phenyltrimethicone, or a mixture thereof.
 5. The power shunt of claim 1, wherein the shaft collar comprises: a set screw configured to allow the load shaft to slide through the shaft collar when the set screw is loosened such that a desired operational distance can be established as the adjustable operational distance from the drive turbine to the load turbine; and the set screw further configured to secure the shaft collar to the load shaft such that the desired operational distance is maintained as the adjustable operational distance from the drive turbine to the load turbine.
 6. The power shunt of claim 1, wherein the rotary array of drive teeth further comprises: the arranged gear pattern of the rotary array of drive teeth configured to provide: a first range of rotary output power to the drive shaft in response to the first range of shear rates of the STF in the chamber resulting from a first range of rotary power from the load device that causes the decreasing viscosity, and a second range of rotary output power to the drive shaft in response to the second range of shear rates of the STF in the chamber resulting from a second range of rotary power from the load device that causes the increasing viscosity, wherein the second range of rotary output power is greater than the first range of rotary output power.
 7. The power shunt of claim 1, wherein the rotary array of load teeth further comprises: the arranged gear pattern of the rotary array of load teeth configured to provide: the decreasing viscosity in response to the first range of shear rates of the STF in the chamber in response to a first range of rotary power from the load device causing first level revolutions per the unit of time of the load shaft to be greater than the revolutions per the unit of time of the drive shaft, and the increasing viscosity in response to the second range of shear rates of the STF in the chamber in response to a second range of rotary power from the load device, wherein the second range of rotary power is greater than the first range of rotary power causing second revolutions per the unit of time of the load shaft to be greater than the revolutions per the unit of time of the drive shaft such that the second revolutions per the unit of time of the load shaft is less than the first revolutions per the unit of time of the load shaft.
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